Hybrid TF-IDF + ABTT Dense Score Fusion

Step 4 — Combining lexical precision with semantic recall

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1. Objective

Steps 2–3 established two complementary similarity signals:

- **TF-IDF** captures *lexical* discriminators: bill identifiers (`1610/מ`, `NNNN/פ`), law names, and domain-specific terminology. High precision on exact-match topics, brittle to paraphrase.
- **ABTT-corrected dense** (AlephBERT $k = 2$, `e5` $k = 1$) captures distributional/semantic similarity after projecting out the register bias. Robust to paraphrase and surface variation, but less sharp on entity-centric topics.

This step tests whether combining them via **score fusion** yields a strictly better similarity score for the subject-linking task.

2. Method

Given precomputed L2-normalized similarity matrices S_{tfidf} and S_{dense} (both 105×105 on the recurring subset), the hybrid score is:

$$S_{\text{hybrid}}(\alpha) \;=\; \alpha \cdot S_{\text{tfidf}} \;+\; (1-\alpha) \cdot S_{\text{dense}}, \qquad \alpha \in \{0.0, 0.1, \ldots, 1.0\}.$$

$\alpha = 0$ is pure dense; $\alpha = 1$ is pure TF-IDF. Both endpoint cases reproduce the Step 2/3 baselines.

Clustering uses S_{hybrid} directly:

- *Agglomerative*: precomputed distance $= 1 - S_{\text{hybrid}}$, average linkage, $K = 28$.
- *Spectral*: precomputed affinity $= \max(S_{\text{hybrid}}, 0)$, $K = 28$.

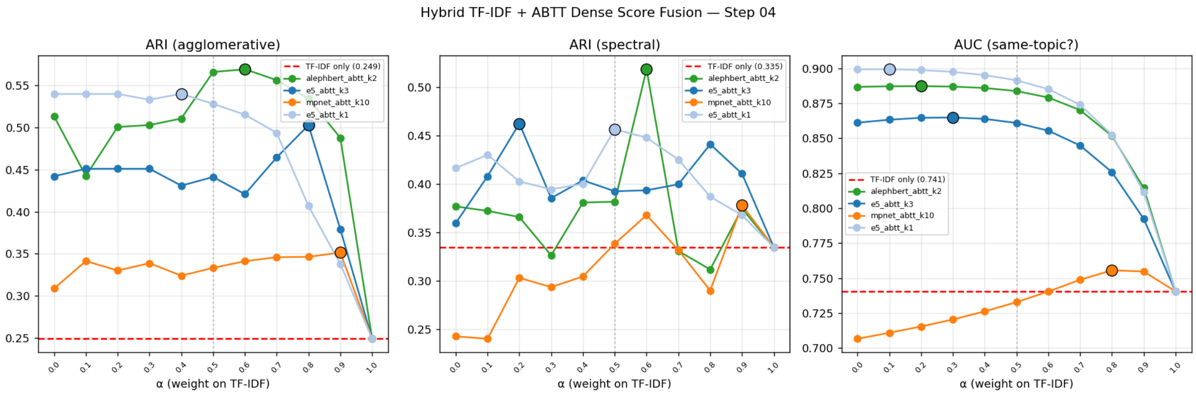
Four ABTT variants are tested: `alephbert_k2`, `e5_k3` (best KMeans in Step 3), `e5_k1` (best agglomerative/AUC in Step 3), and `mpnet_k10`.

3. Results

3.1 Summary table

Configuration	Best α	ARI_{agg}	ARI_{spec}	AUC
TF-IDF only	—	0.249	0.335	0.741
`alephbert_k2`, pure ($\alpha = 0$)	—	0.513	0.377	0.887
`alephbert_k2`, $\alpha = 0.6$	**0.6**	**0.569**	**0.519**	0.879
`e5_k1`, pure ($\alpha = 0$)	—	0.540	0.417	**0.899**
`e5_k1`, $\alpha = 0.4$	0.4	0.540	0.400	0.895
`e5_k3`, $\alpha = 0.8$	0.8	0.503	0.441	0.826
`mpnet_k10`, $\alpha = 0.9$	0.9	0.351	0.378	0.755

Global best: `alephbert_k2` at $\alpha = 0.6$, $\text{ARI}_{\text{agg}} = 0.569$ — a **2.3**× improvement over TF-IDF alone.



KMeans ARI, Agglomerative ARI, and AUC as a function of α for each ABTT variant. Red dashed line = TF-IDF only. Filled circles mark the best α per model. Left end ($\alpha = 0$) = pure dense; right end ($\alpha = 1$) = pure TF-IDF.

3.2 Model-specific behaviour

AlephBERT $k = 2$ (best clustering). The hybrid strictly dominates both components: $\alpha = 0.6$ (slightly TF-IDF-weighted) gives $\text{ARI}_{\text{agg}} = 0.569$ vs. 0.513 for pure dense. The two signals carry complementary information: AlephBERT captures committee-specific paraphrase; TF-IDF captures bill numbers that identify the exact legislative matter.

`e5` $k = 1$ (best AUC). The α curve is essentially flat from 0 to 0.4 ($\text{ARI}_{\text{agg}} \approx 0.540$), suggesting that `e5` $k = 1$ has already internalized the lexical discriminators to the extent that TF-IDF adds no additional information. AUC = 0.899 throughout. **This model is the natural choice for the Task-2 similarity baseline.**

mpnet. Performance improves marginally with TF-IDF mixing but never approaches AlephBERT or e5. mpnet is excluded from further steps.

3.3 AUC is maximized by pure dense

Adding TF-IDF degrades AUC monotonically (TF-IDF's own similarity gap is only 0.069 vs. dense ≈0.34). For *ranking* purposes (retrieval pre-filter for the LLM verifier), pure dense with `e5` $k = 1$ or `alephbert` $k = 2$ is optimal.

4. Takeaways

- **Score fusion helps AlephBERT but not e5.** The difference reflects the models' handling of entity-centric signal: AlephBERT, trained on raw Hebrew text, benefits from the explicit lexical overlay; e5, contrastively trained, absorbs that signal on its own.
- **Two operating points for downstream use:** `alephbert_k2` at $\alpha = 0.6$ for clustering ($\text{ARI}_{\text{agg}} = 0.569$); `e5_k1` pure dense for ranking/AUC (0.899).
- **Score fusion is a zero-cost improvement**: no retraining, no new data, linear combination of precomputed similarity matrices.

5. Future Directions

1. **Similarity threshold baseline (Task 2a).** Use the best representations to set an operating point for the online LINK/NEW decision and characterize the precision–recall trade-off under the anti-duplication constraint. (Step 5.)

2. **LLM verifier (Task 2b).** Use `e5_k1` as a retrieval pre-filter (top-K candidates) and a prompted LLM as verifier.
3. **Fine-grained α sweep** around 0.5–0.7 for AlephBERT (0.05 steps) to pinpoint the optimal mixing weight.
4. **Entity-weighted TF-IDF.** Give extra weight to bill-ID tokens (NNNN/מ) and named entities identified by an NER model to sharpen the lexical component.

Artifacts (this step): `outputs/hybrid_results.json`, `outputs/hybrid_curves.png`, `outputs/best_sim_matrix.npy` (`alephbert_k2`, $\alpha = 0.6$).
Code: `code/hybrid_eval.py`.